23. Graph the following functions

(a)

$$y = \begin{cases} 8, & -4 < x \leq -1 \\ 5, & -1 < x \leq 0 \\ 2, & 0 < x \leq 3 \\ 4, & 3 < x \leq 4 \end{cases}$$

(b)

$$y = \begin{cases} x + 1, & -7 \leq x < -4 \\ -3, & -4 \leq x < 0 \\ 2x - 3, & 0 \leq x \leq 5 \end{cases}$$

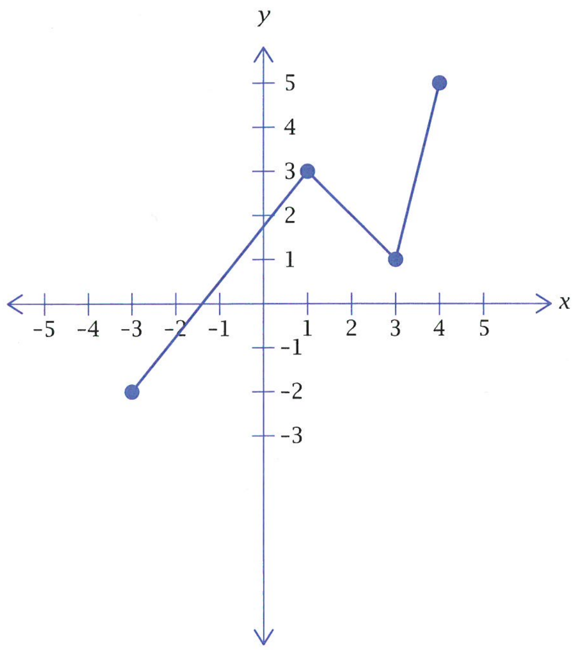
(c)

$$y = \begin{cases} 2 - x, & -3 \leq x \leq -1 \\ x - 2, & -1 < x \leq 2 \\ 2 - x, & 2 < x \leq 3 \end{cases}$$

(d)

$$y = \begin{cases} 5, & x > 3 \\ x, & 0 < x \leq 3 \\ -x - 1, & -4 < x \leq 0 \\ x + 2, & x \leq -4 \end{cases}$$

24. Write an equation for the piece-wise function drawn below:

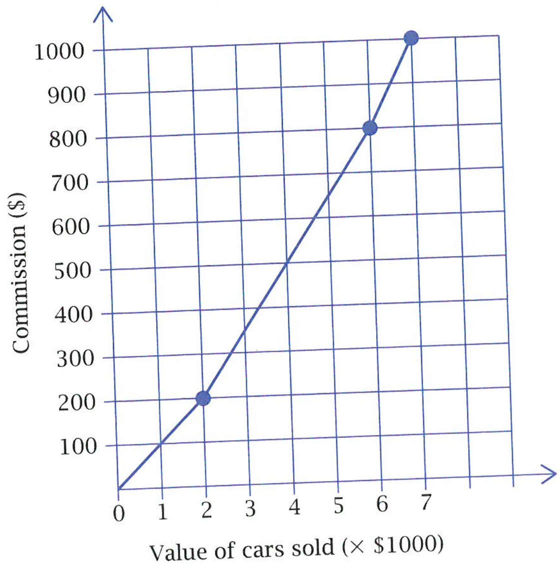


25. John receives a commission for selling second hand cars.

He receives:

- 10% of the first $2000
- 15% of the next $4000
- 20% of any remaining amount.

The piece-wise graph below represents John's commission on the value of cars he sells.



(a) How much commission did John earn for selling cars worth

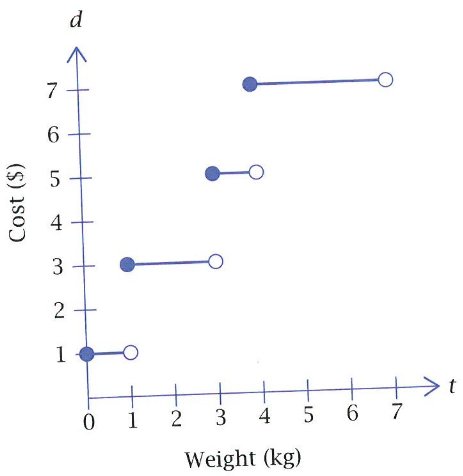
(i) $1500

(ii) $4000

(iii) $6500

(b) John earns a total of $800 commission. What was the value of his car sales?

26. The cost of sending a parcel via post varies according to its weight as shown by the step graph below.



(a) Determine the cost of sending a parcel if it weighs:

(i) 500g

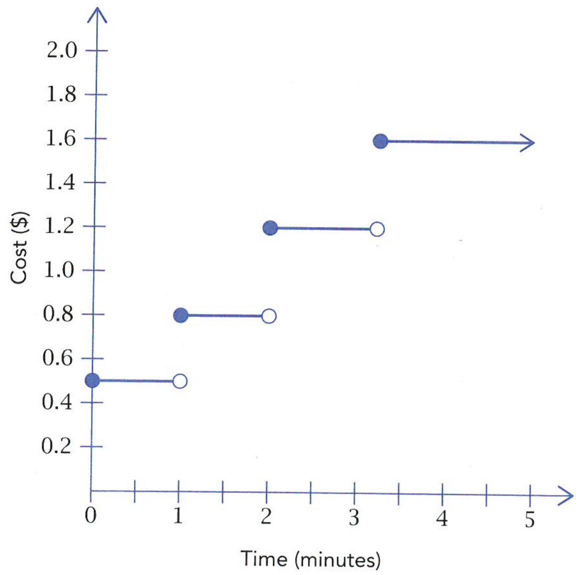
(ii) 1kg

(iii) 3.5kg

(iv) 6.8kg

(b) If a parcel costs $5 to send, determine its possible weight.

27. The cost of mobile phone calls depends on the duration of the call. The step graph below illustrates one mobile provider's cost of mobile calls.



(a) Determine the cost for a mobile phone call which lasts:

(i) 45 seconds

(ii) 2 minutes

(iii) 3 mins 45 seconds

(b) Determine the length of a phone call which costs $1.20.

28. The distance from Derek's house to his friend Peter's house is 80km. Derek leaves one morning for Peter's house at 8.00 am, travelling for 30 minutes at 40 km/h. He stops for ½ hour to buy some supplies.

He leaves the shops and arrives at Peter's house at 10.30 am. The boys study for 1 hour. Derek then travels home at a steady speed arriving at 12.15 pm.

(a) Draw a piece-wise linear graph to represent Derek's journey.

(b) Calculate Derek's average speed from:

(i) 9 am to 10.30 am

(ii) 11.30 am to 12.15 pm

(c) What is the fastest part of the journey? How is this represented on the graph?